Vascular growth, cardiac niches and CM maturation

What the two broad contexts mean

The composition-only K2 solution separates stromal/mural-rich neighborhoods from CM/EC-rich myocardial neighborhoods. These are recurring neighbor mixtures, not two new cell types, exhaustive anatomical compartments, or a reproduction of the atlas paper’s expression-based12 niches.

What cell concentration means

Smooth-muscle, epicardial and immune self-contacts, and early fibroblast–immune contacts, exceed local abundance-matched expectations with both native and refined boundaries. This supports spatial organization; it does not identify a secreted signal or prove an effect on CM maturation.

How sprouting and maturation can coordinate

The2021 paper establishes perinatal expansion from a pre-existing coronary plexus and DLL4/NOTCH feedback restraint. The2026 atlas finds positive P7 vascular-proximity/maturation associations and proposes PTPRM contact signaling. Vessel growth and subsequent integration can support CM maturation without a universal negative local sprouting–maturity relationship.

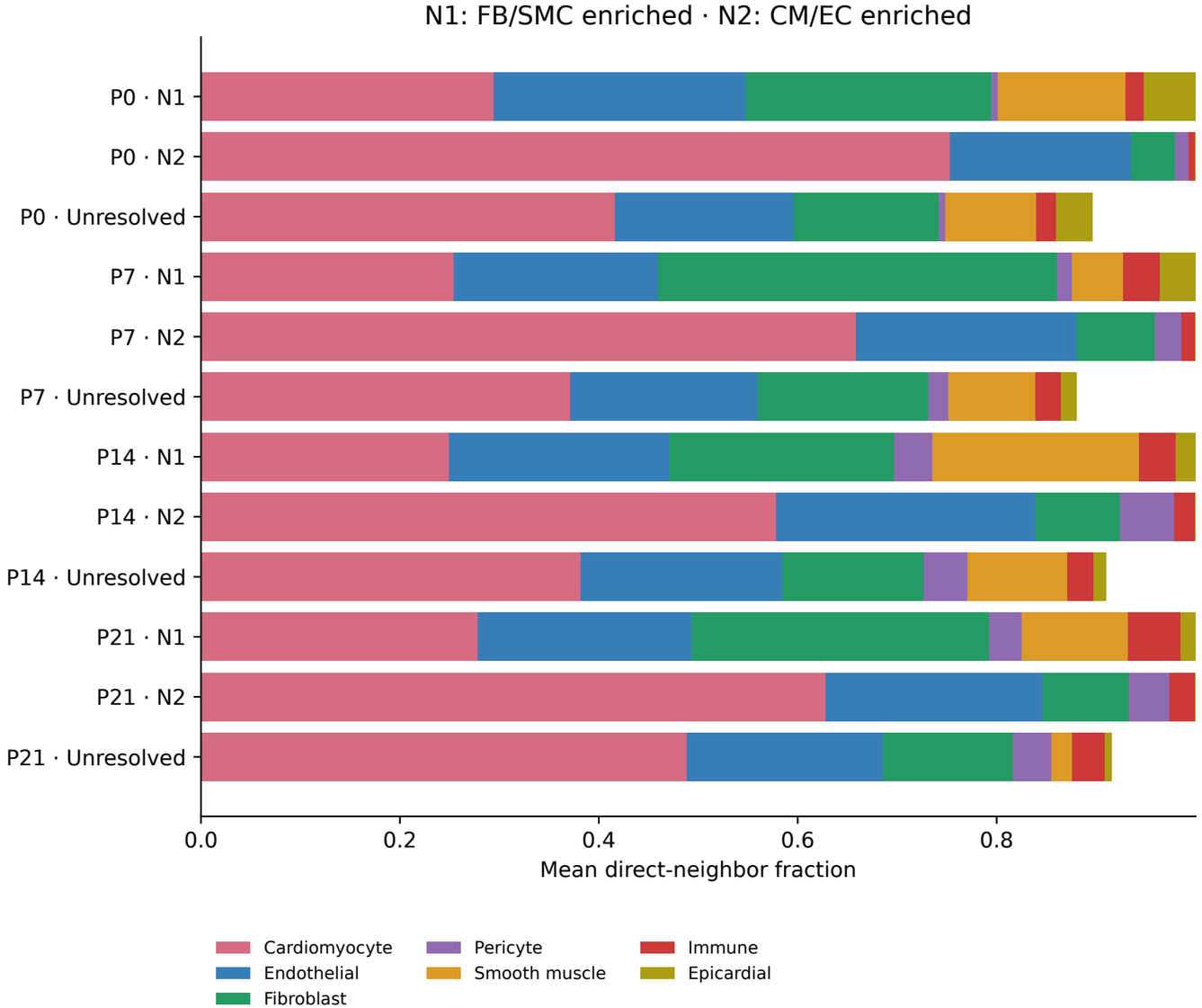
What this extension adds

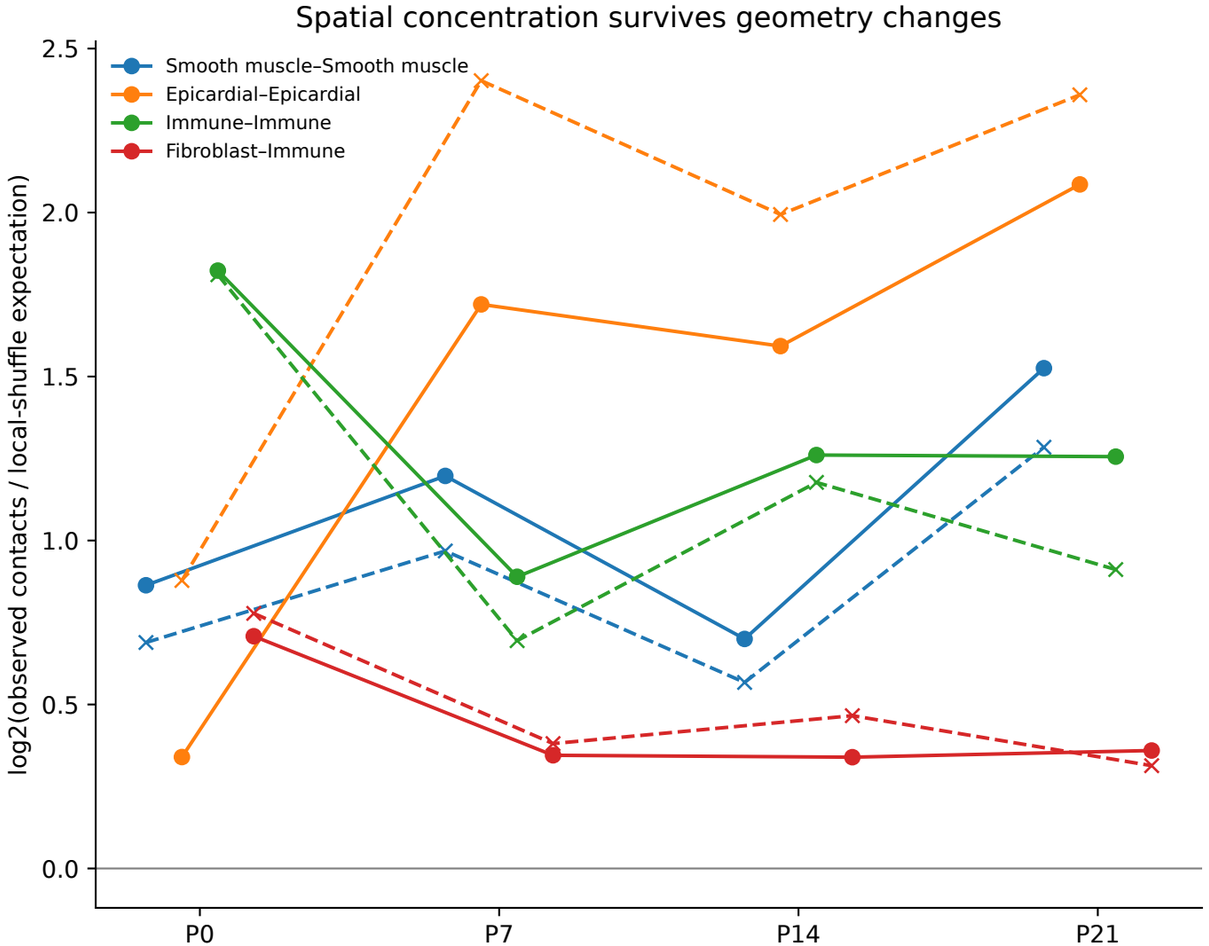
Measured coronary/endocardial/Notch proxies; whole-RNA library support and coherence; native-polygon and refined-boundary comparisons; graph rings with censor flags; source/receiver controls; source-selection audits; and an actually trained, held-out Wormhole benchmark. Unmeasured sprout transcripts are never invented.

Interpretation boundary

One spatial section per age confounds age with specimen/section. RNA expression is not oxygenation, perfusion, verified sprouting or signaling activation. Associations and cross-sectional state correspondences do not establish causal influence or track developmental ancestry.

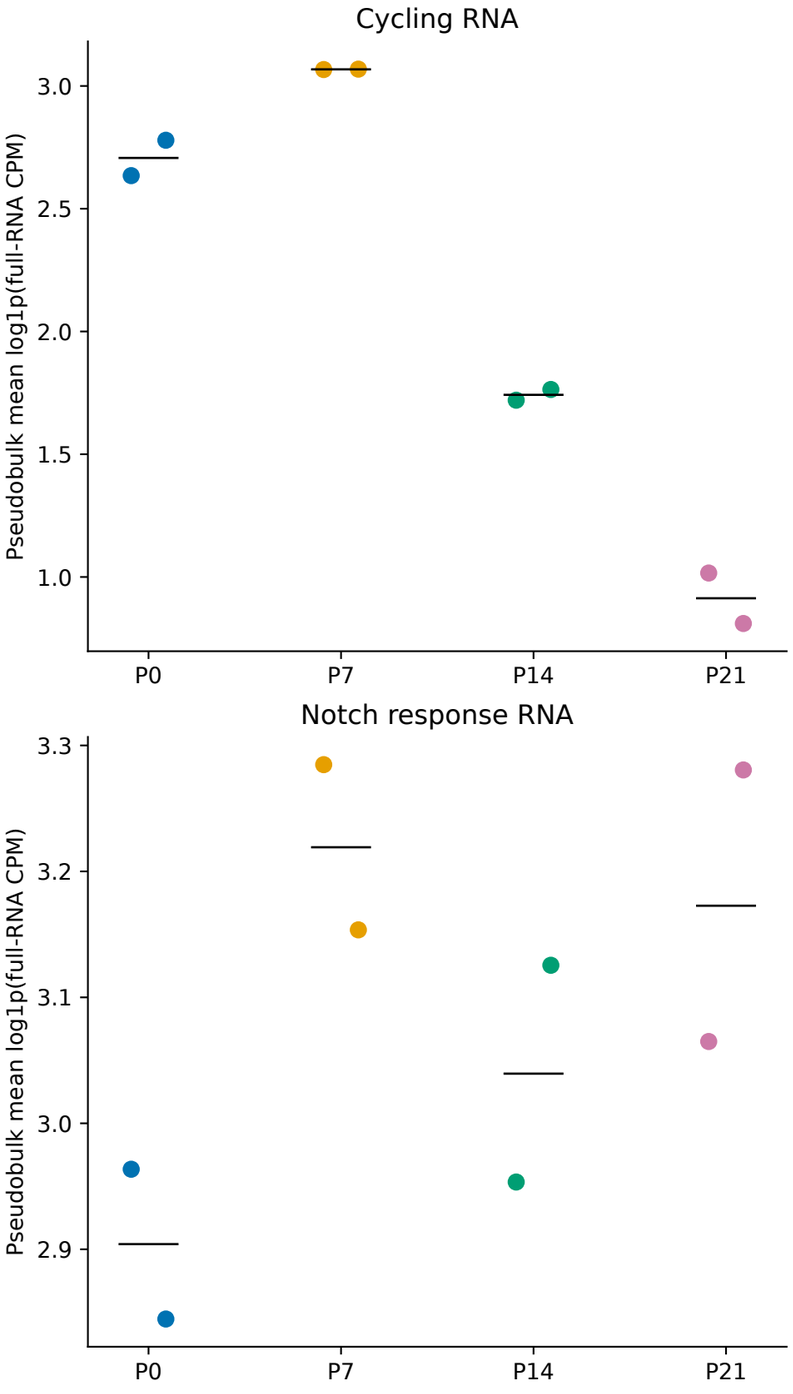
Broad contexts and contact concentration

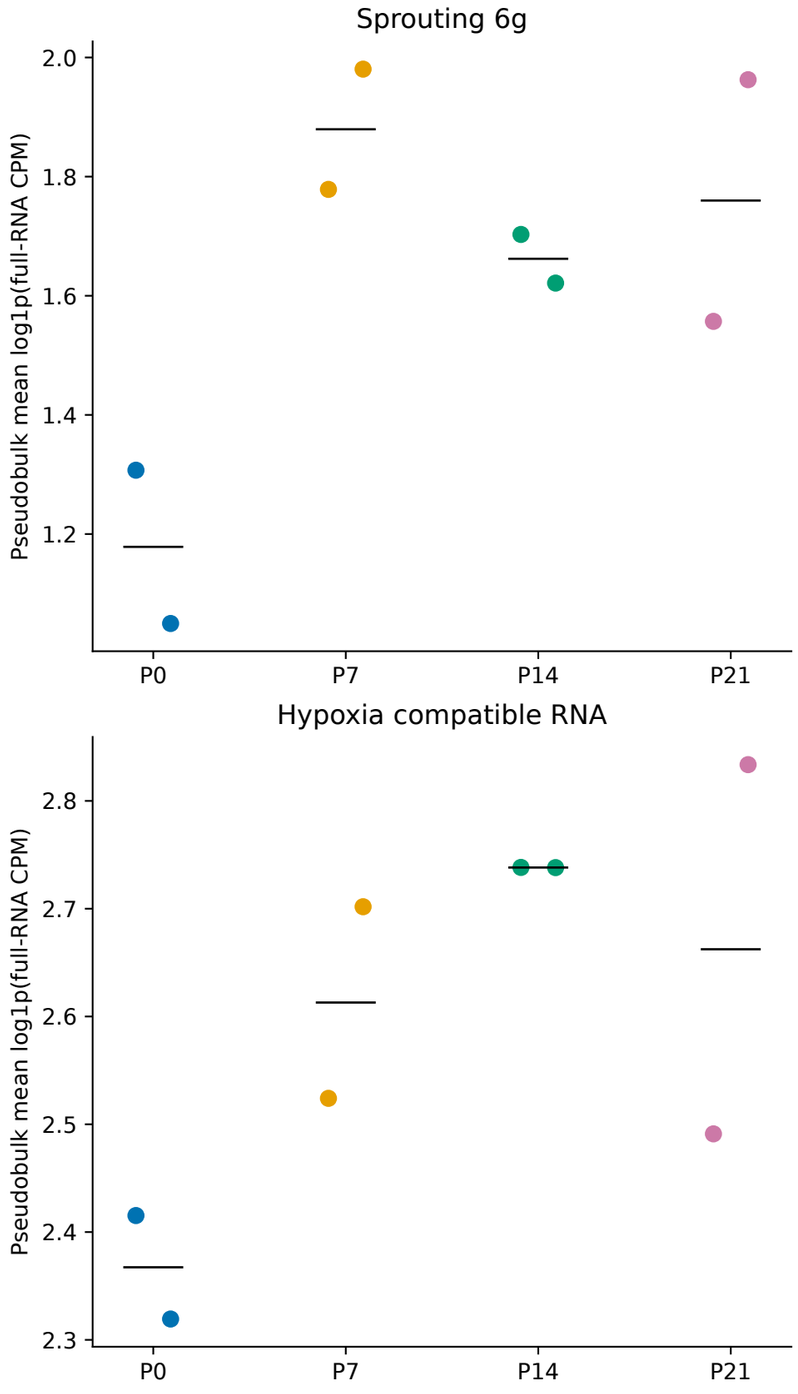


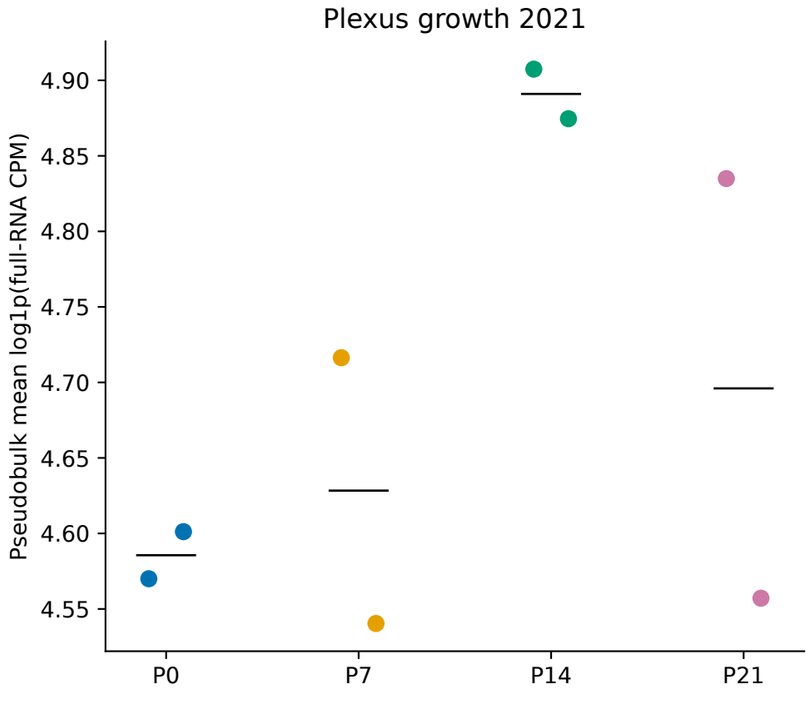


Solid circles: native polygons within1µm; dashed crosses: refined shared boundary. Null shuffles labels within200µm blocks. Association, not signaling.

Whole-transcriptome snRNA-seq: cycling is better supported than one sprouting module



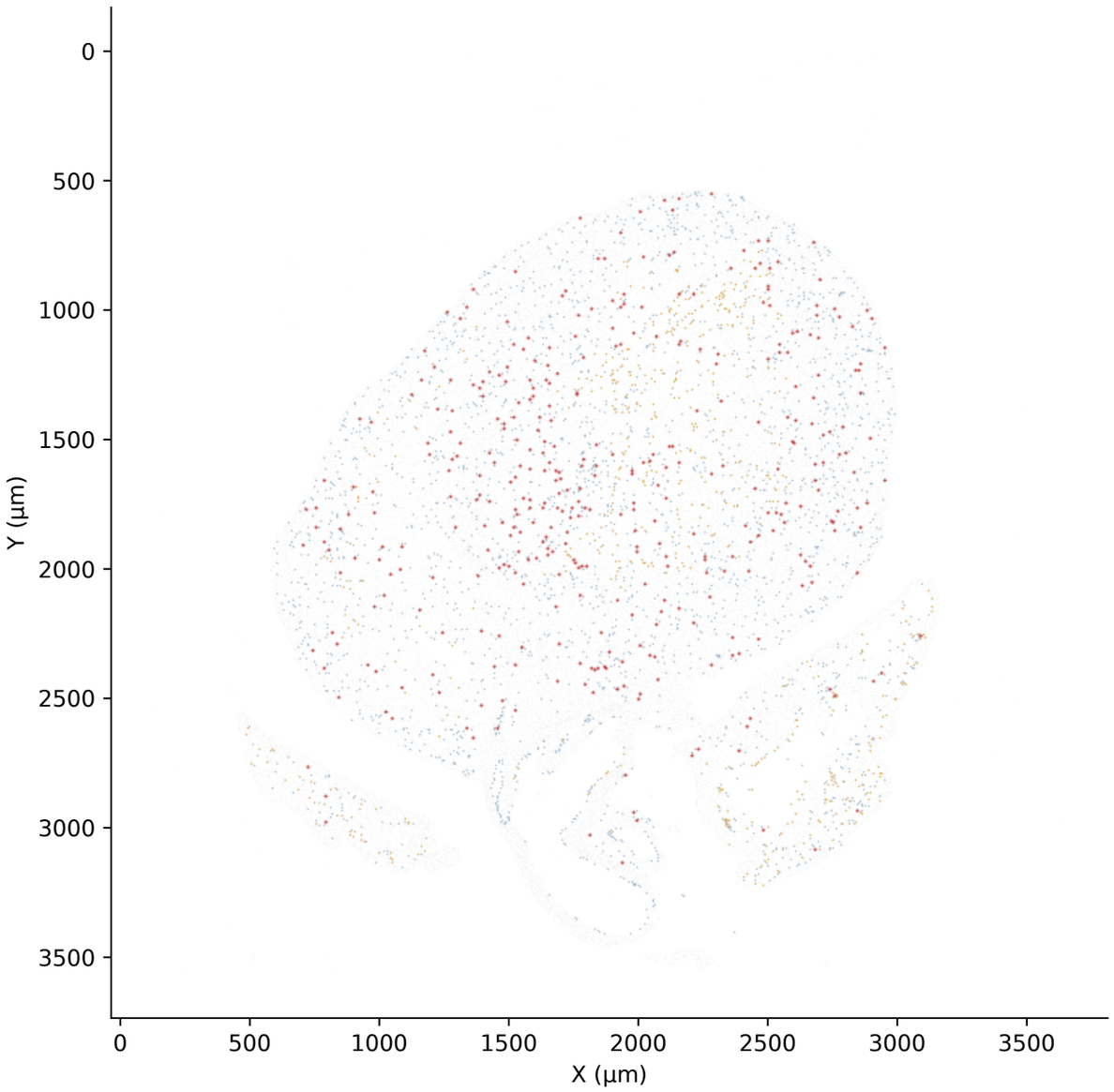




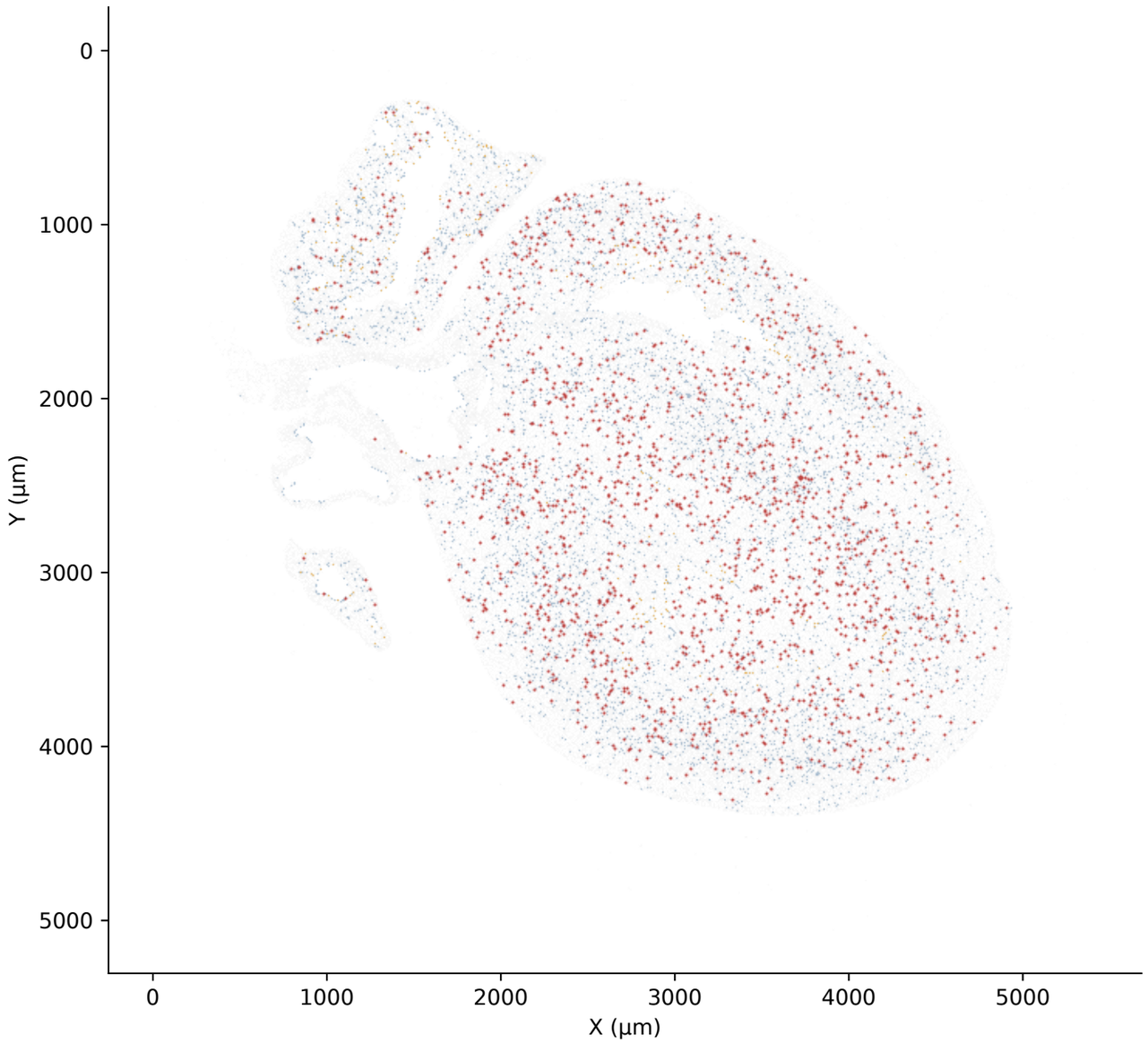
Two libraries/age, corrected counts. Six-gene sprouting panel is sparse; scores do not establish sprout morphology. NA: <2 sufficiently detected genes.

Other eligible EC Endocardial-compatible Growth-compatible

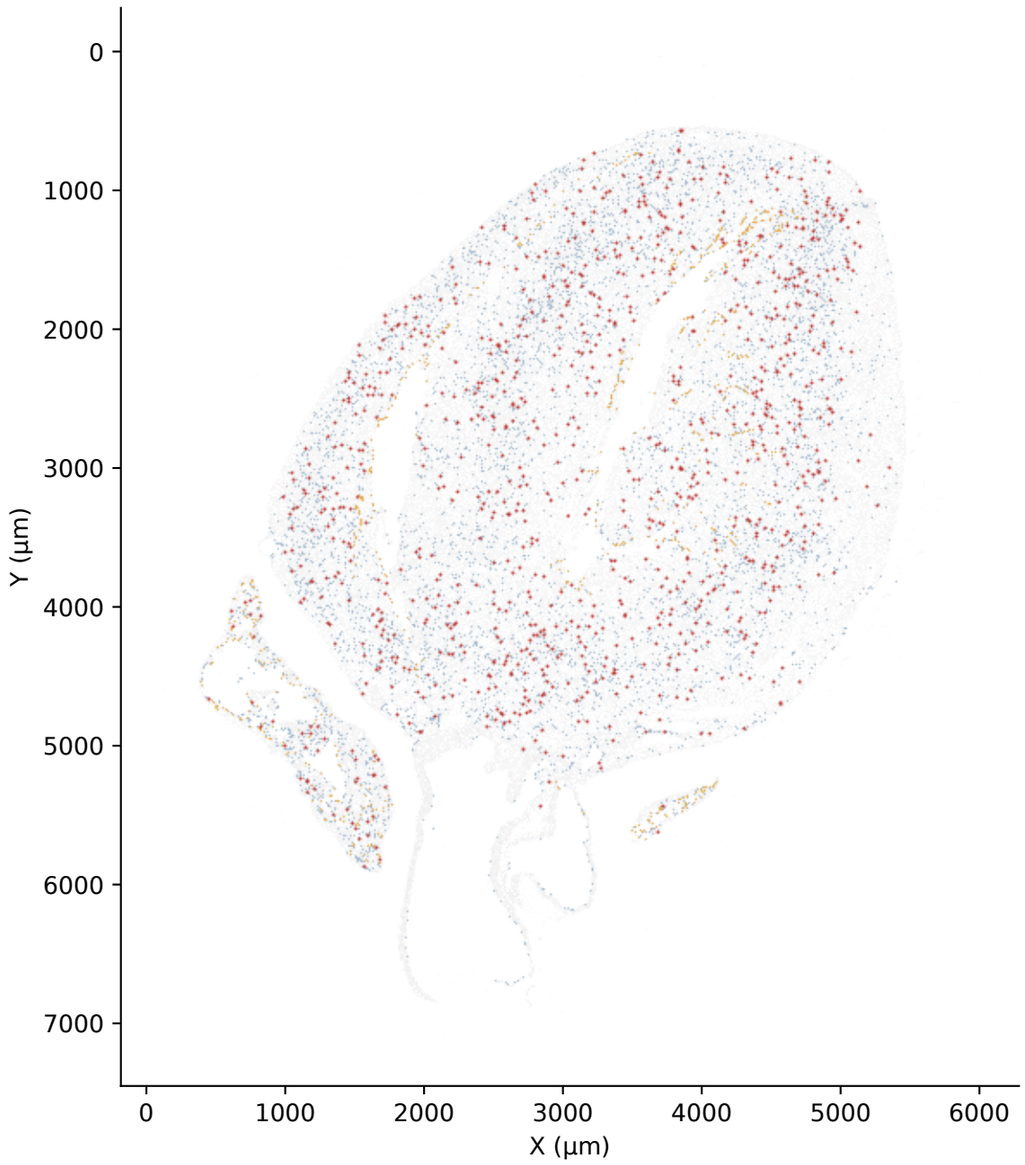
P0: 383 growth-compatible / 3,732 eligible ECs



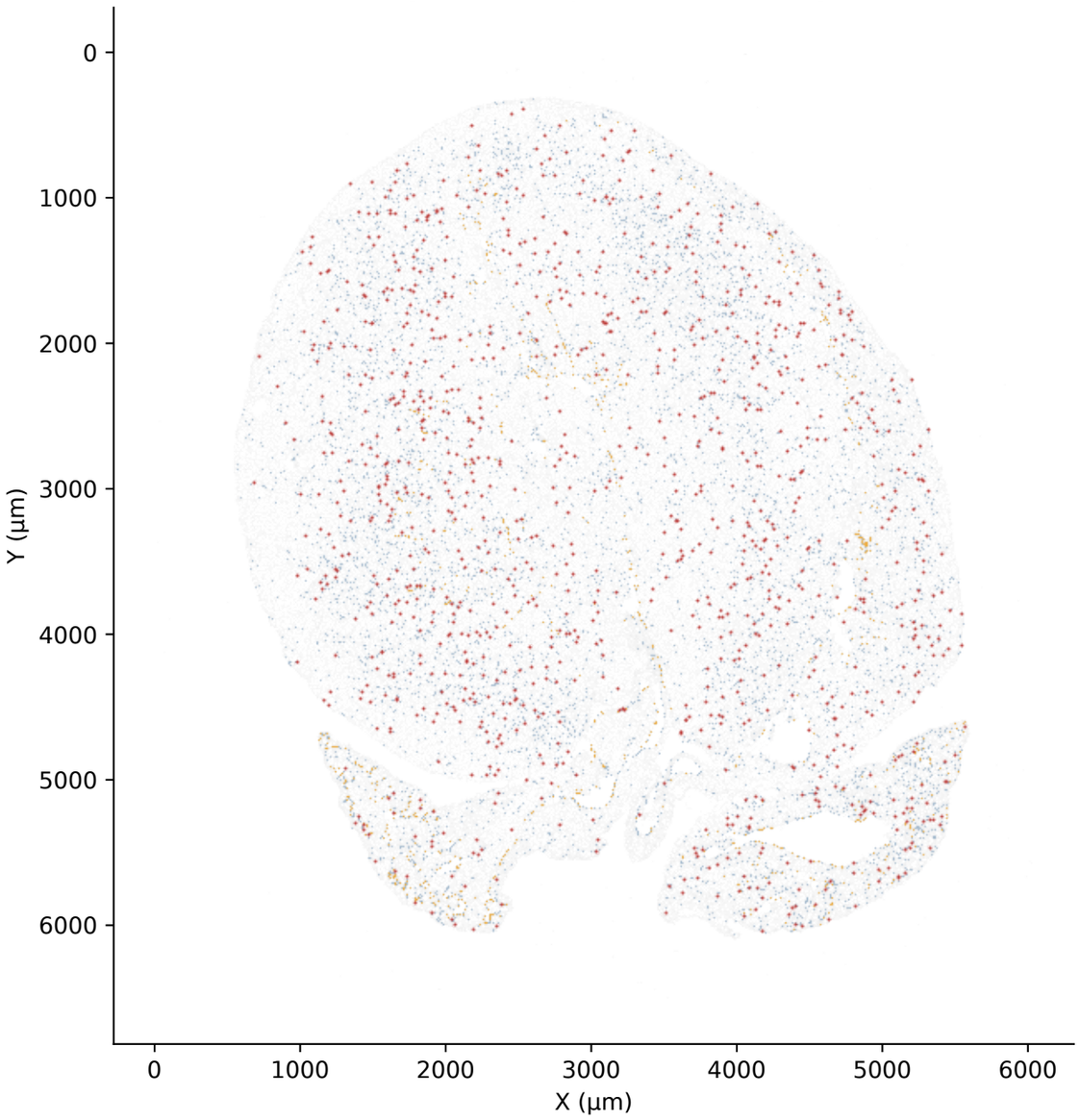
P7: 1,664 growth-compatible / 8,944 eligible ECs



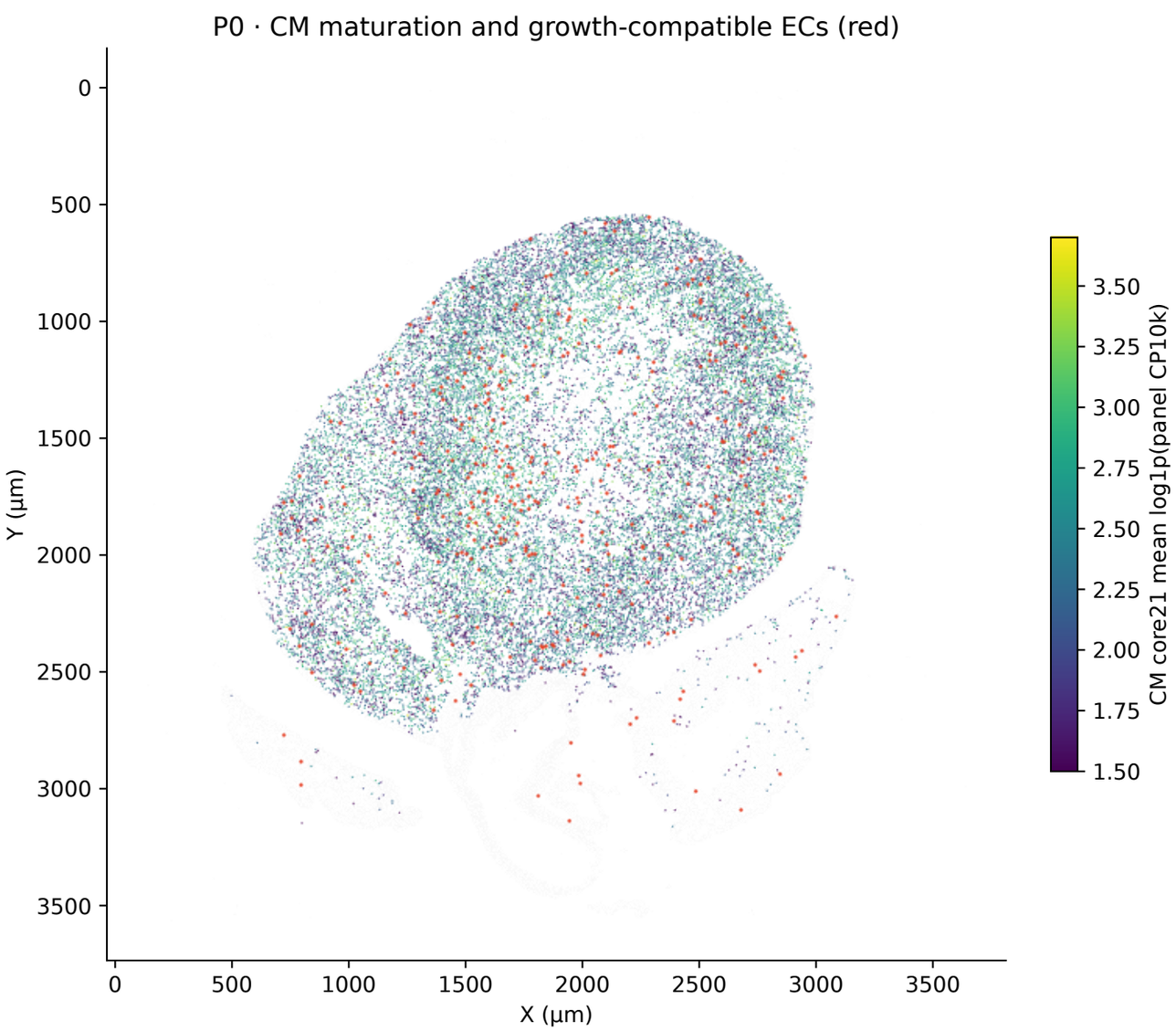
P14: 972 growth-compatible / 8,086 eligible ECs

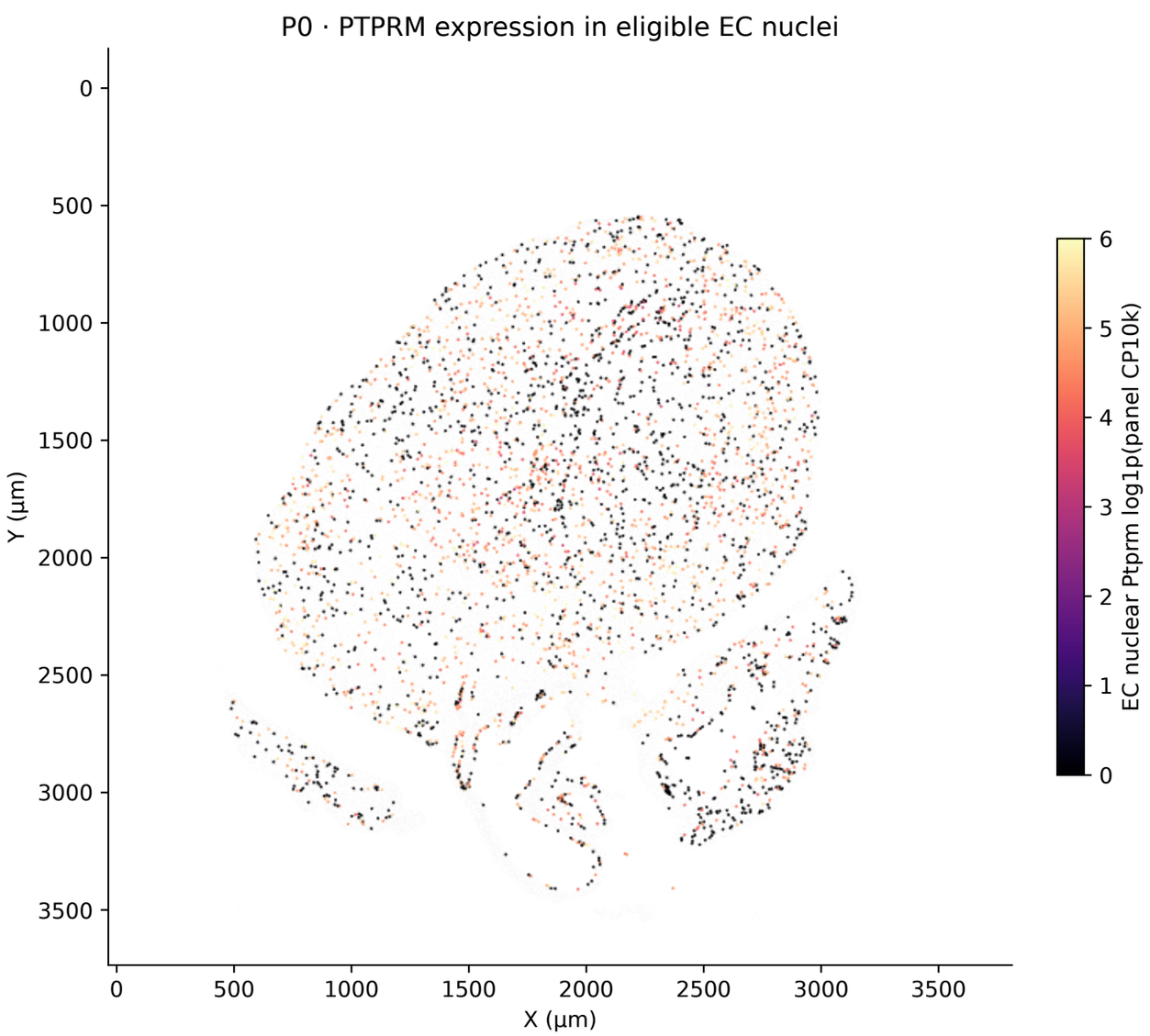


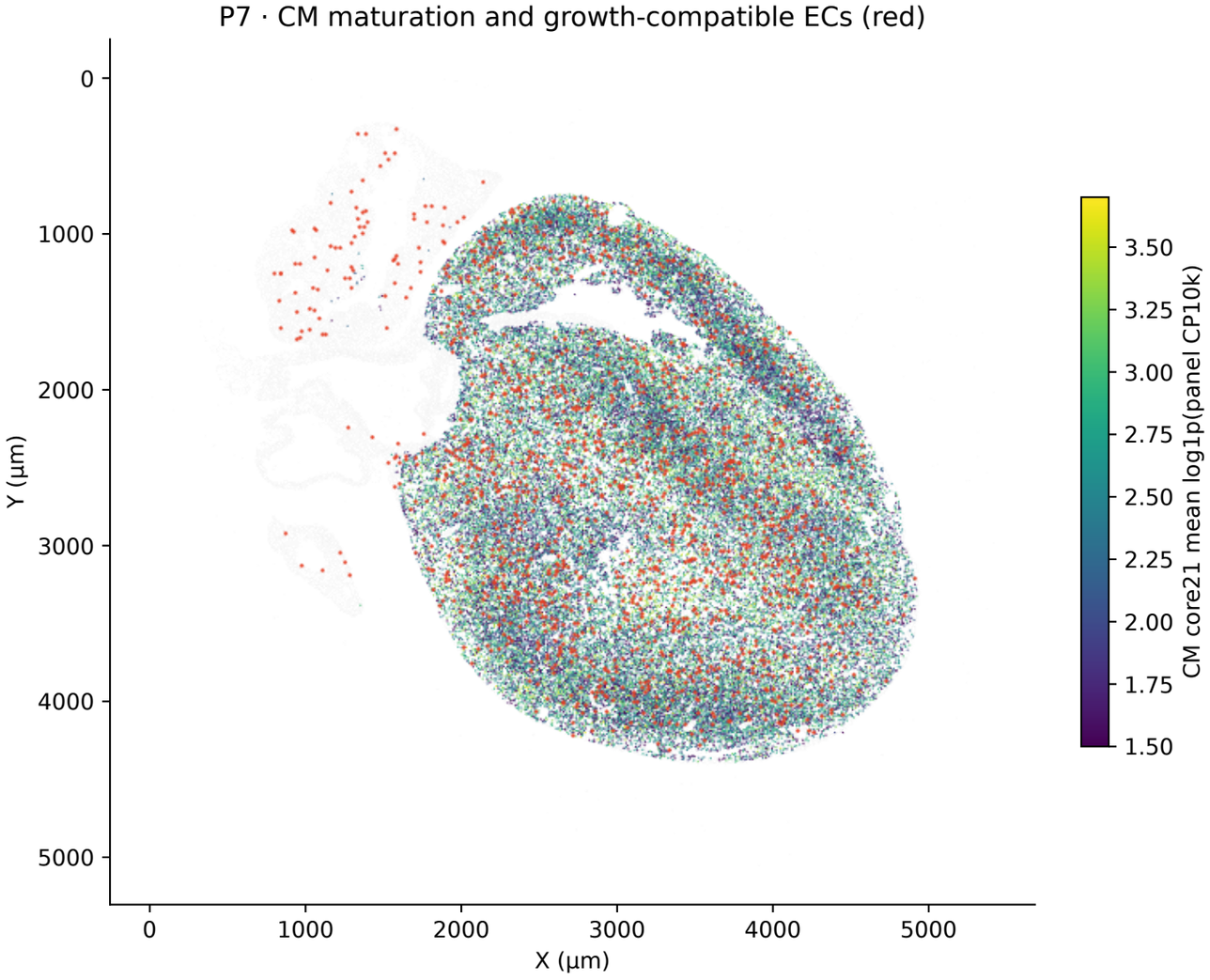
P21: 1,079 growth-compatible / 8,601 eligible ECs

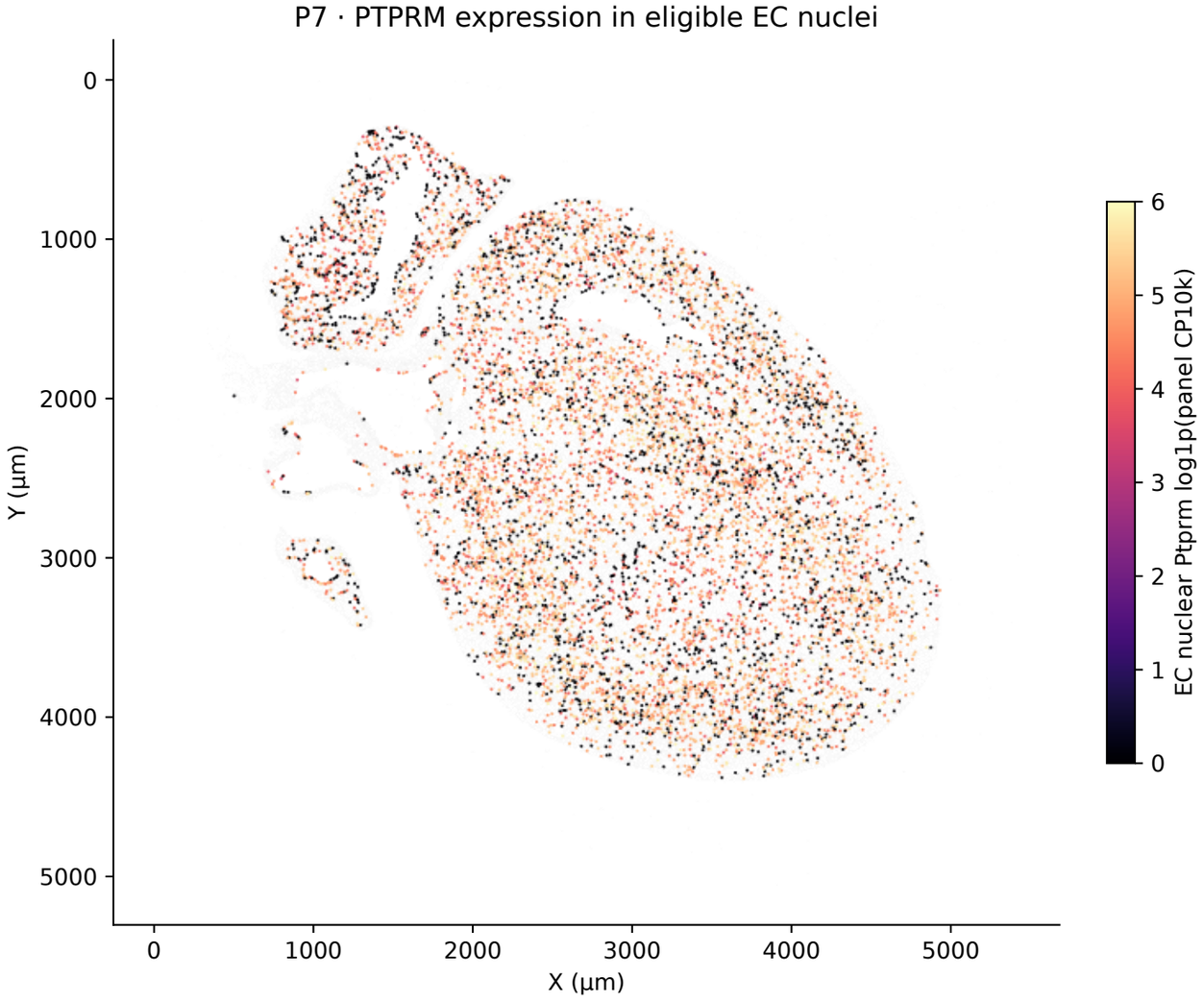


Growth-compatible = primary EC, not assigned endocardial/lymphatic identity, Flt4 AND Dll4 core co-detection. Unresolved subtypes remain possible; no verified sprouts or oxygenation.

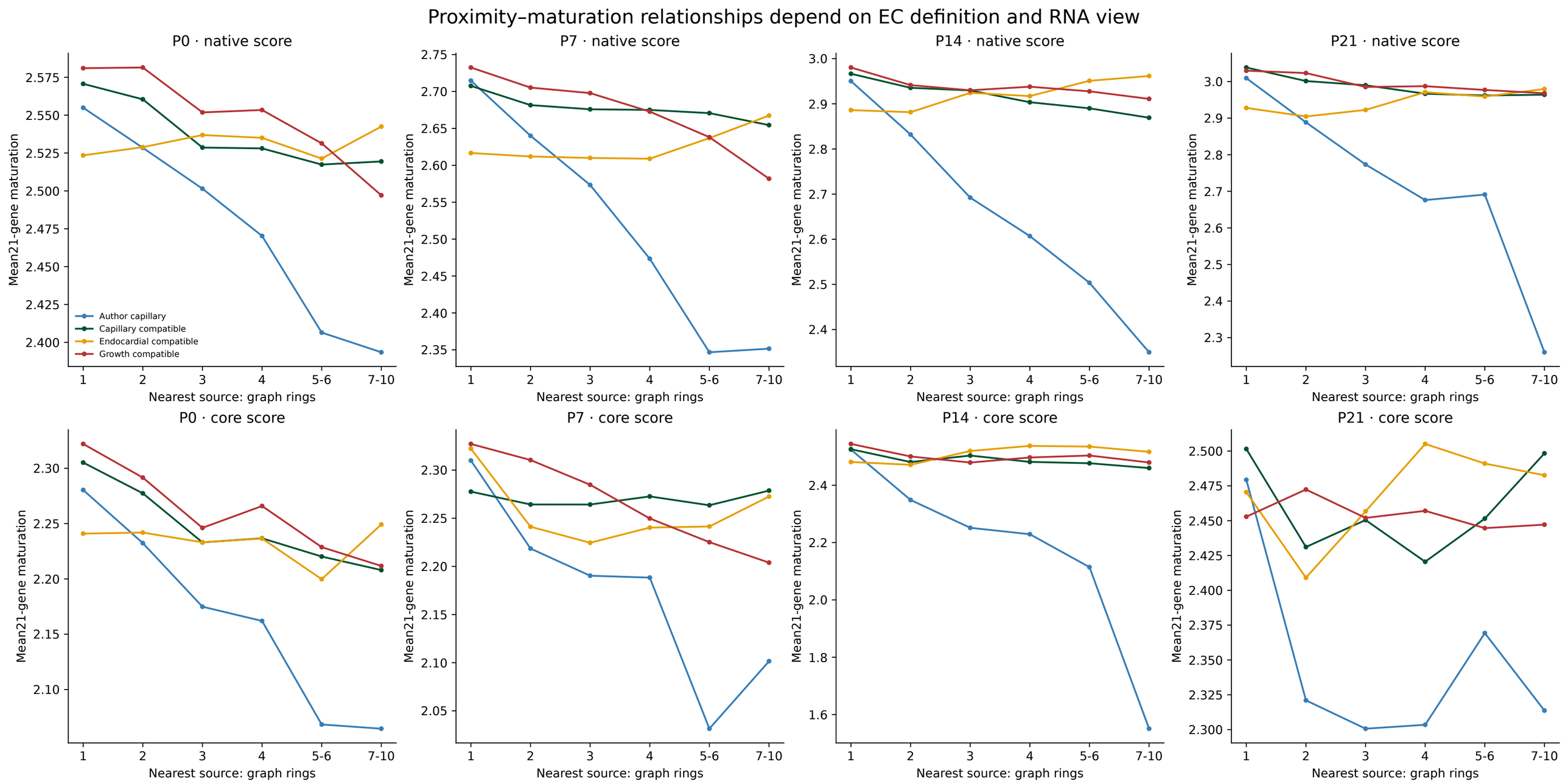


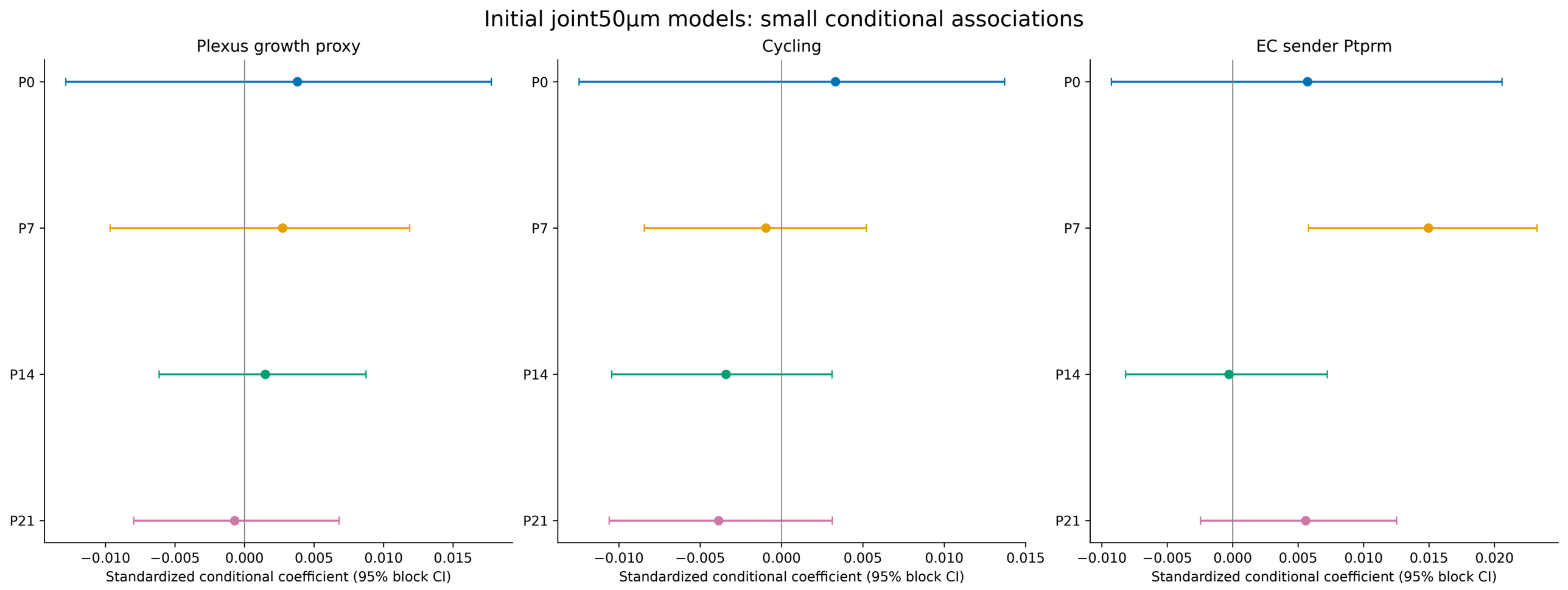




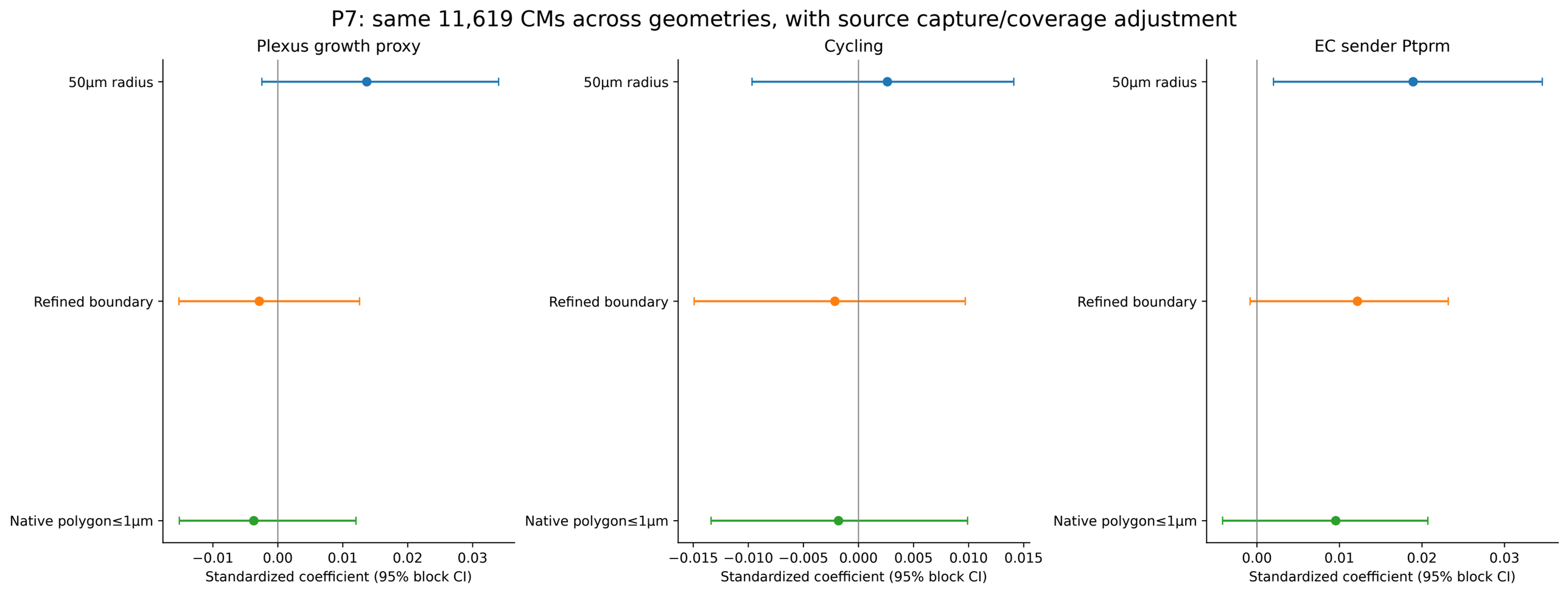


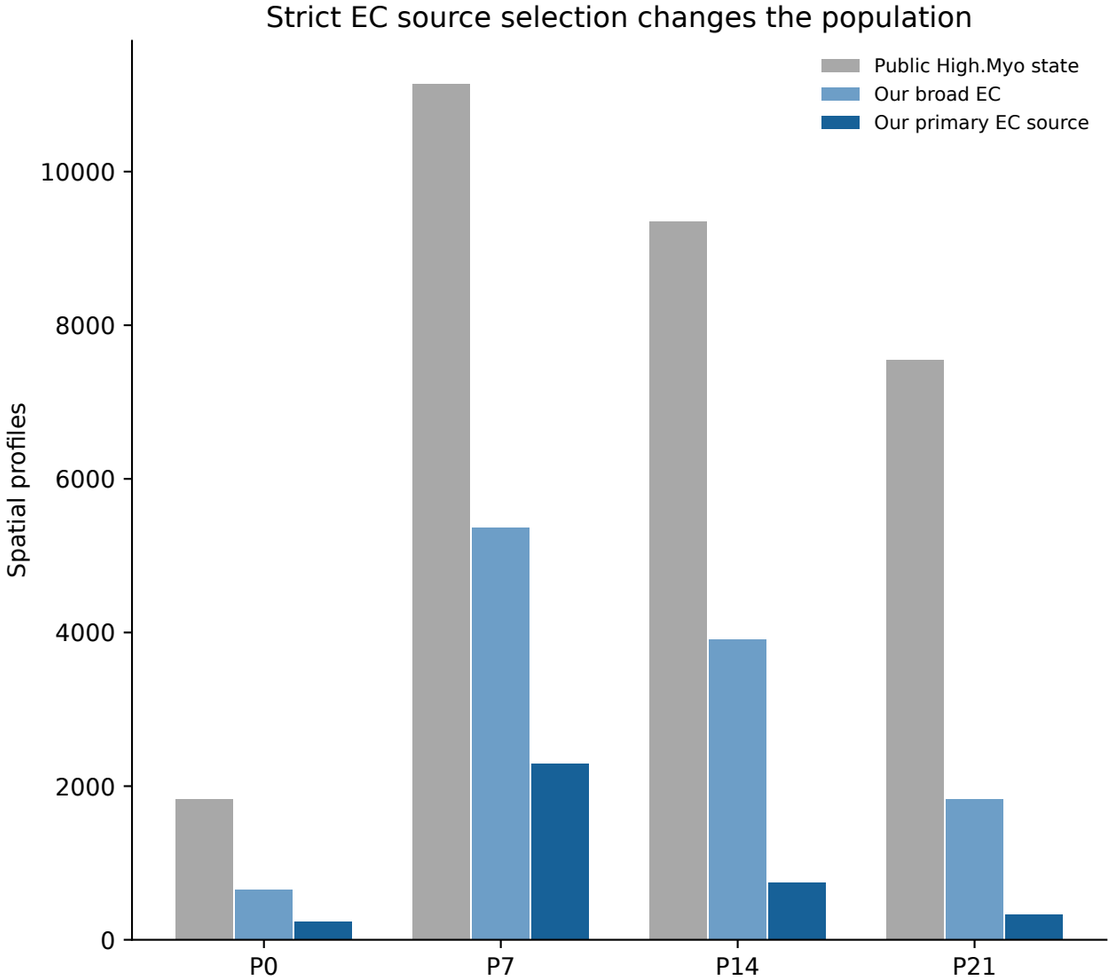
Spatial overlap is descriptive. Adjacent sections/ages are different specimens; these maps do not measure EC-specific causal signals or local oxygen delivery.

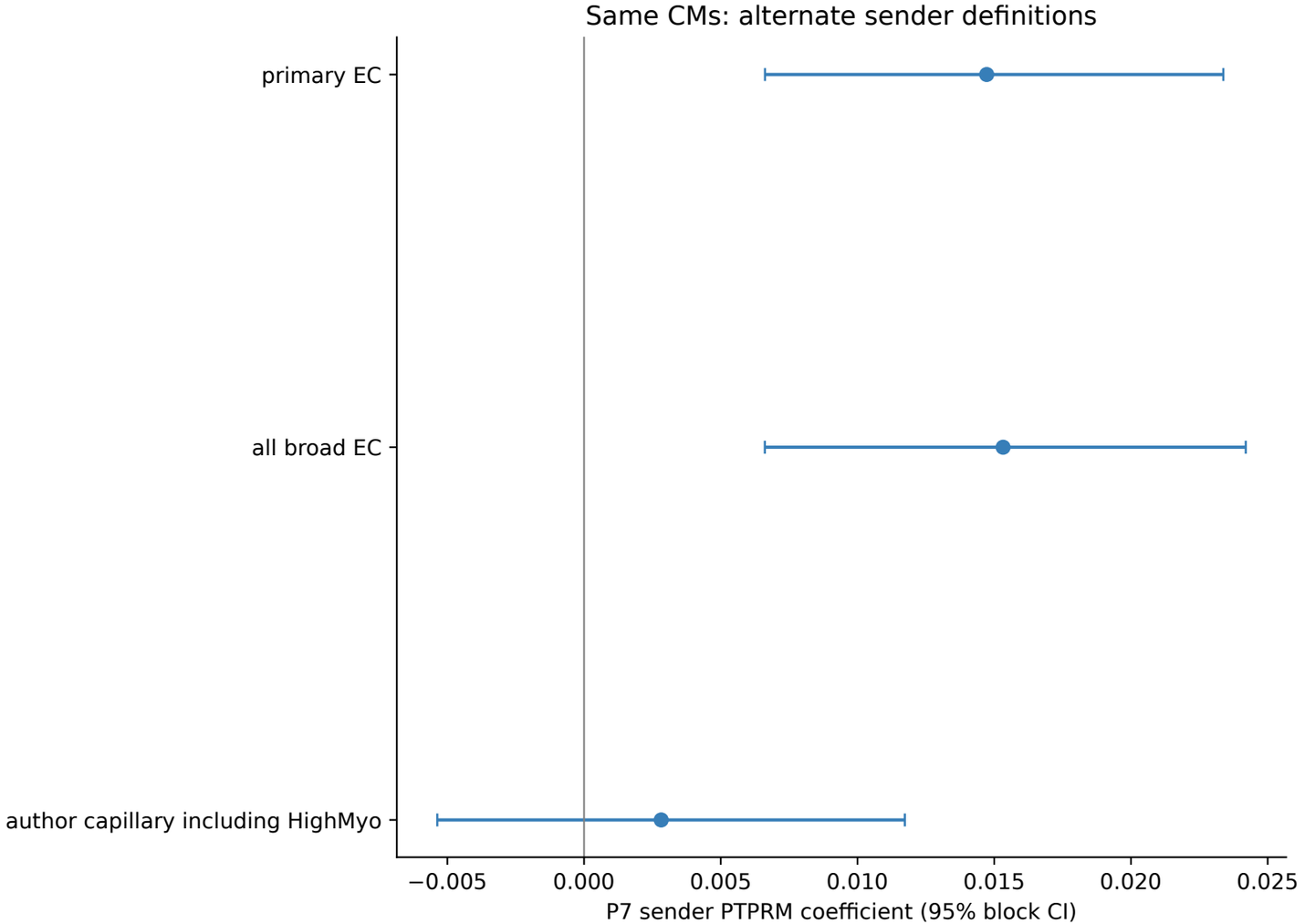

Proximity–maturation relationships depend on EC definition and RNA view
P0 · native score
P7 · native score
P14 · native score
P21 · native score
P0 · core score
P7 · core score
P14 · core score
P21 · core score
Mean21-gene maturation
Nearest source: graph rings
1
2
3
4
5-6
7-10
Author capillary
Capillary compatible
Endocardial compatible
Growth compatible
Pruned Delaunay graph, reached CMs only; censored/unreachable profiles are tabulated separately. Descriptive curves, not exact author graph reproduction.



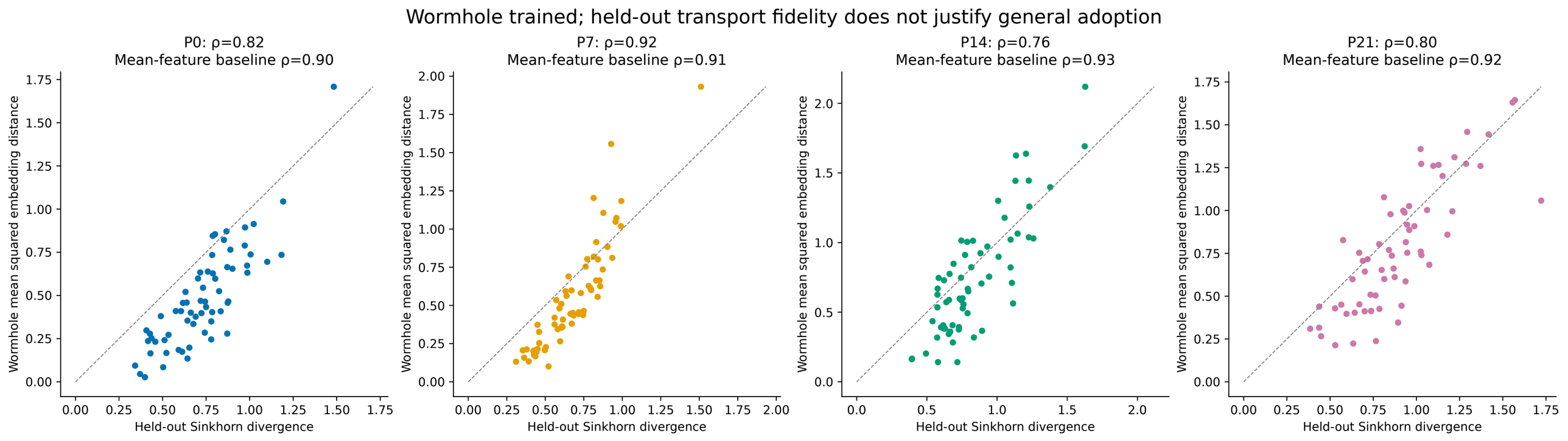
Joint EC programs/families, CM sender/receiver, focal capture/area, other-cell fractions, density, spatial splines and distance proxies. No causal attribution.

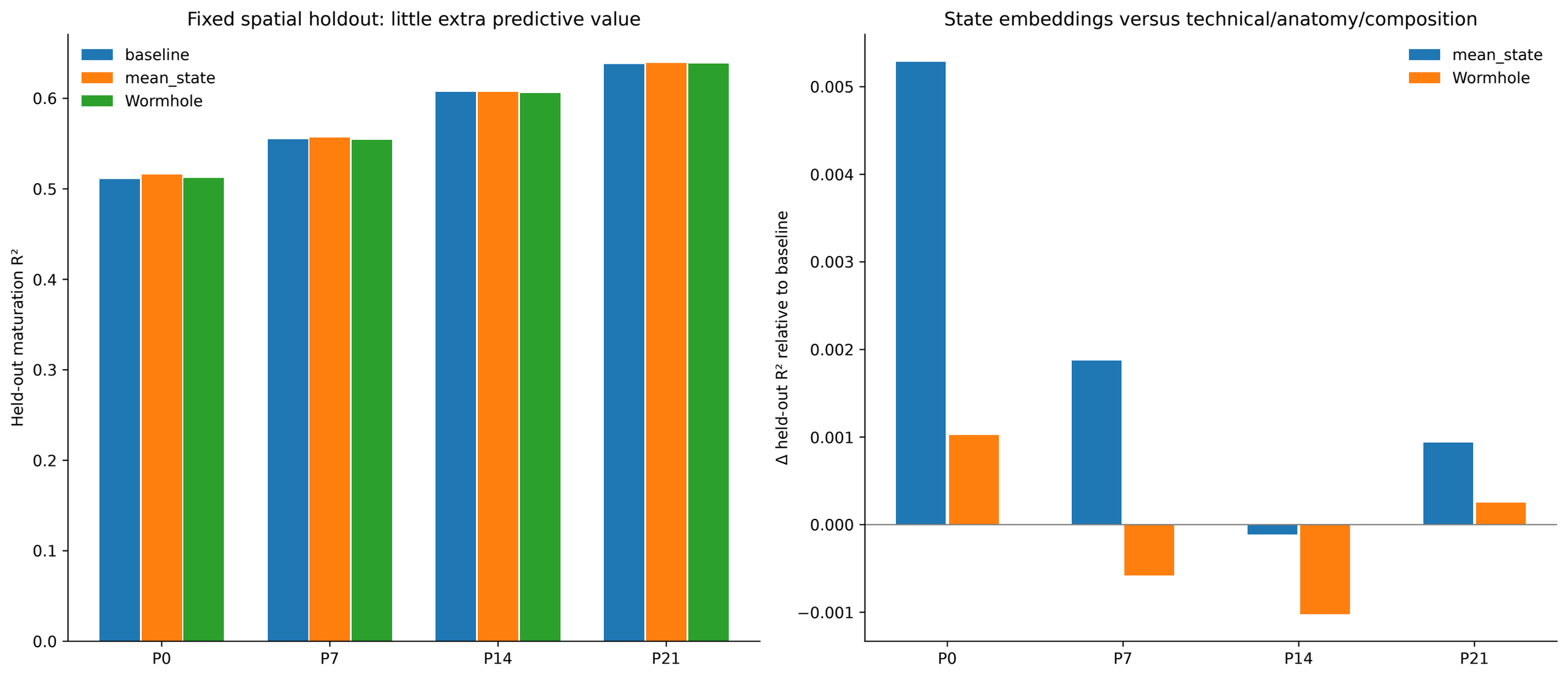

P7: same 11,619 CMs across geometries, with source capture/coverage adjustment
Plexus growth proxy
Cycling
EC sender Ptprm
50µm radius
Refined boundary
Native polygon≤1µm
−0.01
0.00
0.01
0.02
0.03
−0.015
−0.010
−0.005
0.000
0.005
0.010
0.015
Standardized coefficient (95% block CI)
Same focal CMs; neighboring EC nuclear/core depth and source-selection coverage included. Different graphs define different exposure variables.



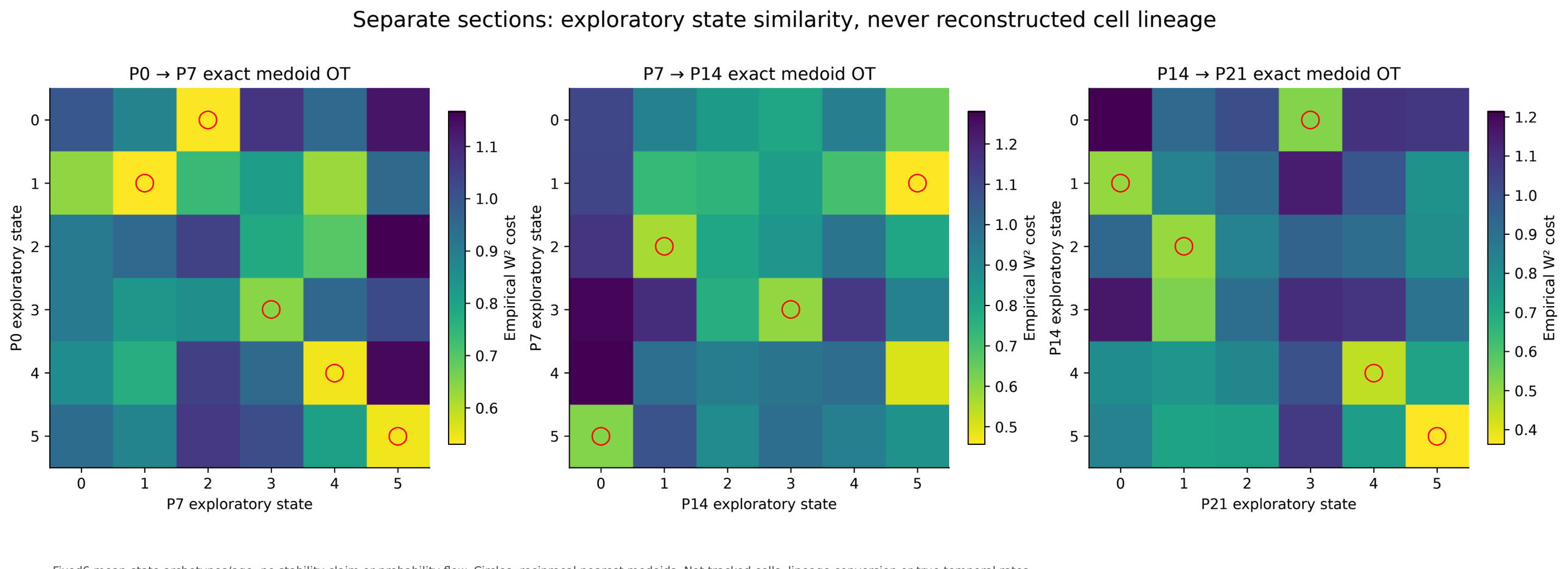


Public High.Myo may contain biological CM–EC programs and mixed boundaries. Exclusion is not proof of doublets; inclusion is not independent identity validation.


Wormhole trained; held-out transport fidelity does not justify general adoption
P0: ρ=0.82
Mean-feature baseline ρ=0.90
P7: ρ=0.92
Mean-feature baseline ρ=0.91
P14: ρ=0.76
Mean-feature baseline ρ=0.93
P21: ρ=0.80
Mean-feature baseline ρ=0.92
Wormhole mean squared embedding distance
Held-out Sinkhorn divergence
64 reserved within-age pairs/stage, non-CM gene-state clouds≤32 cells; actual encoder/decoder1200 steps. CPU benchmark, not default paper architecture.



150µm train/test buffer; sources in test clouds excluded from PCA fitting; focal maturation genes excluded from non-CM expression and normalization.


Separate sections: exploratory state similarity, never reconstructed cell lineage
P0 → P7 exact medoid OT
P0 exploratory state
P7 exploratory state
Empirical W² cost
P7 → P14 exact medoid OT
P7 exploratory state
P14 exploratory state
Empirical W² cost
P14 → P21 exact medoid OT
P14 exploratory state
P21 exploratory state
Empirical W² cost
Fixed6 mean-state archetypes/age; no stability claim or probability flow. Circles: reciprocal nearest medoids. Not tracked cells, lineage conversion or true temporal rates.

What is supported, and what needs direct validation

Supported spatial structure

Broad neighbor mixtures and selected contact enrichments survive changes in segmentation geometry. The measured growth-compatible EC proxy is more frequent among eligible ECs at P7 than P0; it remains a panel-defined state, not an angiogenesis rate.

EC–CM maturation interpretation

The small positive P7 EC-sender PTPRM association in50µm models has intervals including zero for paired native/refined boundaries and approaches zero using the public capillary/High.Myo sender definition. It is a candidate association, not an independently established EC-specific mechanism. Growth/cycling neighborhoods do not consistently show lower overall CM maturation at bothP0 andP7.

Method selection

Wormhole was trained on3200 source-cell clouds, assessed on800 regional held-out clouds, and benchmarked against exact/smoothed empirical transport and simple means. It fails the all-age metric gate and adds little held-out maturation prediction. Mean-state features/exact medoid transport remain descriptive comparisons; no true niche dynamics or new stable6-state taxonomy is claimed.

Next experiments

Use independent hearts/sections and anatomically registered inner/outer myocardium; quantify vascular branches and EC EdU alongside the CM maturation panel; validate candidate EC–CM boundaries by imaging; compare CM-directed and EC-directed perturbations to resolve sender necessity. The2021 paper’s Hypoxyprobe and lineage-tracing evidence cannot be replaced by sparse RNA scores.

Sources

Lu etal., Nature Cell Biology23,967–977(2021), doi:10.1038/s41556-021-00747-1, full uploaded13-page PDF read. Wang etal., Spatially guided in vivo single-cell functional genomics of postnatal heart, doi:10.1038/s44161-026-00861-z, full PMC13561863 text/XML read. See paper_review_and_pipeline_changes.tsv for figure/method-specific implications.